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fp-tools: A Reproducible Command-Line Platform for Classical and Multiscale ATAC-seq Footprinting, Supervised TF-Binding Prediction, Motif Discovery, and Variant Scoring

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Abstract

(1) Background: Transcription-factor (TF) footprinting from ATAC-seq is central to inferring regulatory activity, but the surrounding ecosystem is fragmented across single-purpose tools with heterogeneous interfaces, inconsistent reproducibility, and uneven test coverage, which raises the cost of building end-to-end regulatory-genomics analyses. (2) Methods: We present `fp-tools`, a standalone, pip-installable Python package that unifies bias correction, footprint scoring, differential TF-binding detection, and aggregate visualization behind stable command-line entry points, with optional YAML and graphical wrappers. Around this core we add explicitly opt-in scientific modules: multiscale and nucleosome-aware scoring, motif-centric supervised TF-binding prediction, motif-relaxed and motif-free candidate recovery, de novo motif discovery orchestration, footprint-aware variant scoring, single-cell pseudobulk aggregation, replicate-aware differential-binding uncertainty reporting, and a competition-aware decomposition of overlapping TF-scale and nucleosome-scale footprint signals. (3) Results: The package provides deterministic, regression-tested workflows; a tiered public-data benchmark design with AUROC, AUPRC, recall-at-FDR, calibration, and bootstrap-confidence-interval metrics; and paper-ready figure generators. In ENCODE chromosome-4 benchmarks spanning three cell lines and 12 cell–TF tasks, an integrated `fp-tools` score combining accessibility, motif strength, GC content, and available cut-site footprint features achieved the highest AUROC in every task (mean AUROC 0.80), outperforming motif-only scoring (0.77), raw accessibility (0.52), and footprint-only scoring where available (0.55). The same benchmark shows that footprint occupancy is recoverable only at Tn5 cut-site resolution (AUROC 0.68) and not from smoothed coverage (0.24). New modules preserve the behavior of the classical commands when they are not invoked. (4) Conclusions: `fp-tools` integrates established footprinting strengths with modern multiscale, supervised, and variant-oriented extensions in a single reproducible package, lowering the barrier to standardized, benchmarkable ATAC-seq regulatory analysis.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; supervised TF-binding prediction; multiscale footprints; variant scoring; reproducibility; command-line bioinformatics

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1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of the active regulatory landscape of a cell [1]. Within accessible regions, sequence-specific transcription factors (TFs) protect the DNA they occupy from transposition, leaving characteristic depletions—"footprints"—flanked by hyper-accessible edges. Detecting and interpreting these footprints supports inference of TF occupancy, differential regulation between conditions, and the functional consequences of non-coding variation.

A mature but *fragmented* ecosystem of methods addresses pieces of this problem. Classical footprinting and differential-binding frameworks such as TOBIAS [2] and HINT-ATAC [3] correct enzymatic sequence bias and score footprints at a single TF-relevant scale. Multiscale and nucleosome-aware approaches, exemplified by the PRINT/scPrinter family [4], model accessibility structure across a range of footprint widths. Supervised TF-binding-site (TFBS) predictors such as maxATAC [5] and sequence-to-profile models such as ChromBPNet [6] learn occupancy from labeled data or raw sequence, and downstream interpretation tools and variant scorers translate models into motif and allele-level effects. Each tool typically ships its own interface, input conventions, and assumptions, so assembling a coherent, reproducible pipeline—and *benchmarking* alternatives on common ground—remains laborious and error-prone.

Three practical gaps motivate this work. First, **interface fragmentation**: moving from bias correction to footprint scoring to differential binding to visualization commonly spans several packages with incompatible I/O. Second, **reproducibility and testing**: many footprinting workflows lack deterministic regression tests, pinned environments, and machine-checkable output contracts, making it hard to know whether a change altered results. Third, **extensibility without regression**: adding new scientific behavior (e.g., multiscale scoring or supervised prediction) often forces users onto entirely new tools rather than extending a stable base.

We address these gaps with `fp-tools`, a standalone Python package that (i) provides stable, primary command-line entry points for the four core workflows—bias correction, footprint scoring, differential TF-binding detection, and aggregate plotting; (ii) layers optional YAML- and GUI-based wrappers without displacing the command line; and (iii) introduces a series of *explicitly opt-in* scientific modules that extend the platform while leaving the classical commands unchanged when they are not invoked. The guiding principle is conservative: keep existing bulk ATAC workflows stable and reproducible, and add novelty as composable modules with their own tests, documentation, and benchmark hooks. This paper describes the package architecture, the new modules, the tiered benchmarking and figure-generation design, and the reproducibility engineering that ties them together.

2. Materials and Methods

2.1. Package Architecture and Core Workflows

`fp-tools` is distributed as a pip-installable package (`fp-tools-bio`) with vendored Cython internals for performance-critical signal operations. Core scientific tools live under a single namespace and are exposed through stable console entry points. The four primary commands are:

- `atac-correct` — Tn5 sequence-bias correction producing corrected and expected/bias bigWig tracks and a QC report;
- `score-footprints` — footprint, sum, mean, and pass-through scoring modes over corrected signal within regions;

- `detect-tf-binding` — motif-anchored differential binding across one or more conditions, with repeated-condition replicate grouping and a skewness report for multi-condition runs;
- `plot-aggregate` — aggregate footprint visualization from single BED files or directories, with control overlays, fixed-grid layouts, labels, and CSV exports.

Legacy aliases (`ATACorrect`, `FootprintScores`, `ScoreBigwig`, `BINDetect`, `PlotAggregate`) are retained for backward compatibility. A YAML runner (`fp-tools-run`) expands declarative configs into the same commands, and an optional Streamlit GUI (`fp-tools-gui`) provides forms, run history, and output discovery. Direct CLI usage is the primary interface; YAML and GUI are wrappers that never replace it.

2.2. Opt-in Scientific Modules

New behavior is added as separate modules, each with its own console script, unit tests, and documentation, and each inert unless explicitly invoked:

- *Multiscale and nucleosome-aware scoring*: `score-footprints -score multiscale` computes depletion across a configurable set of scales (default 8–147 bp), with `max/mean` scale collapse, an optional summary bigWig, and a compressed NPZ tensor sidecar (`-output-multiscale-npz`) of shape $n_{\text{scales}} \times$ positions for downstream visualization.
- *Supervised TFBS prediction*: `fp-tools-build-tfbs-features` assembles motif-centric feature tables (genome GC, multiscale/signal summaries, optional overlap labels); `fp-tools-train-tfbs-model` and `fp-tools-predict-tfbs` train and apply tabular models with saved bundles and calibration metrics.
- *Motif-relaxed and motif-free recovery*: `fp-tools-generate-candidates` nominates candidates from score bigWigs and peaks without requiring motif hits, with an optional lower-threshold motif-BED merge path; `fp-tools-rerank-candidates` integrates candidate, motif, family, and model scores into a ranked site table with provenance.
- *De novo motif discovery*: `fp-tools-export-candidate-fasta`, `fp-tools-meme-command`, and `fp-tools-motif-discovery-plan` export candidate-centered FASTA and emit reproducible MEME/DREME, Tomtom, and summary command plans; `fp-tools-summarize-mo` parses MEME/Tomtom output into TSV and HTML reports with inline consensus logos.
- *Footprint-aware variant scoring*: `fp-tools-score-variants` performs genome reference checks, candidate-overlap annotation, deterministic ref/alt sequence-context deltas, optional JASPAR/MEME PWM ref/alt motif deltas, and optional tabular-model probability deltas.
- *Pseudobulk aggregation*: `fp-tools-pseudobulk` groups single-cell ATAC fragments by annotation columns into pseudobulk fragment files, TSV/YAML manifests, and downstream command plans.
- *Replicate-aware differential-binding reporting*: `fp-tools-bindetect-replicate-report` (this work) summarizes a BINDetect `*_results.txt` table into per-comparison effect sizes, p-values, replicate support category, and a p-value–derived uncertainty model (Section 2.3).
- *Competition-aware footprint decomposition*: `fp-tools-decompose-competition` (this work) decomposes multiscale signal into competing TF-scale and nucleosome-scale components (Section 2.4).

2.3. Replicate-aware Differential-Binding Uncertainty

BINDetect-style result tables report, per motif and comparison, an aggregate effect size (`change`) and a two-sided p-value, but not per-replicate scores. We infer replicate counts n_c per condition from (in order of priority) an explicit `condition/n_replicates` or condition/replicate map, <condition>_n_replicates columns in the result table, or a single-replicate fallback. Each comparison is labeled replicate-supported when $\min(n_1, n_2) \geq 2$ and single-replicate otherwise. To attach uncertainty we recover an approximate standard error from the reported p-value, $\mathrm{SE} = |\mathrm{change}|/z$ with $z = \Phi^{-1}(1 - p/2)$, and report a replicate-shrunk effect $\widehat{\mathrm{change}} = w \cdot \mathrm{change}$ with shrinkage weight $w = n_{\min}/(n_{\min} + 1)$ that grows toward 1 with replicate support, together with a 95% interval $\widehat{\mathrm{change}} \pm 1.96\,\mathrm{SE}$. The tool emits a per-comparison long-form table, a comparison-level summary (significant-motif counts, median effect SE, mean shrinkage weight), and an optional diagnostic figure.

2.4. Competition-aware Overlap Decomposition

Overlapping regulatory elements produce footprint signals at multiple scales simultaneously: short TF-scale depletions can coincide with wider nucleosome-scale structure. Using the multiscale NPZ tensor, we partition scales into a TF-scale band (default 3–30 bp) and a nucleosome-scale band (default 120–200 bp), and collapse each band to a positionwise non-negative profile t and n by taking the per-position maximum over band scales. We then decompose the signal additively at each position into shared (competing), TF-only, and nucleosome-only components,

$$\mathrm{shared} = \min(t, n), \quad \mathrm{tf\_only} = t - \mathrm{shared}, \quad \mathrm{nuc\_only} = n - \mathrm{shared},$$

and integrate over the region. A *competition index* is the shared fraction of total signal, $\mathrm{shared}_{\mathrm{AUC}}/(\mathrm{tf\_only}_{\mathrm{AUC}} + \mathrm{nuc\_only}_{\mathrm{AUC}} + \mathrm{shared}_{\mathrm{AUC}})$, and each region is labeled by its dominant component (`tf`, `nucleosome`, `competing`, or `none`). The tool emits per-region and dataset-level tables and an optional figure.

2.5. Public Data, Benchmark Tiers, and Metrics

Benchmarks are organized as versioned TSV manifests recording source, benchmark tier, cell type, donor, TF, assay, accessions, assembly, output type, format, URL, checksum, status, local path, split, and notes. A discovery script queries the ENCODE REST API for released human GRCh38 ATAC-seq and matched TF ChIP-seq/CUT&RUN; a resumable downloader writes per-benchmark download reports with checksums and local paths. Raw and processed public data, and large benchmark outputs, are kept out of version control; scripts, manifests, schemas, tiny fixtures, and figure code are tracked.

The benchmark is deliberately *tiered* rather than monolithic (Table 1): a fast CI-smoke tier on tiny fixtures; a bulk matched tier (ENCODE bulk ATAC vs. matched TF ChIP/CUT&RUN) as the primary classical benchmark; a depth-robustness tier over down-sampled ATAC; a motif-removal tier for motif-relaxed/free and de novo validation; a supervised-TFBS tier; a variant tier; and a later pseudobulk tier. For binary TF-binding labels we compute AUROC and AUPRC globally and per TF-cell task, recall at 1%, 5%, and 10% FDR, precision at fixed recall, calibration curves with expected and maximum calibration error, Brier score for probability models, per-TF macro averages, Wilcoxon signed-rank tests for paired method comparisons, and stratified bootstrap confidence intervals for key deltas. Metric, calibration, and bootstrap computations are implemented as standalone, unit-tested scripts and combined by a benchmark pipeline runner that also emits PDF/SVG/PNG figure panels.

Table 1. Tiered benchmark design. Large data and outputs are untracked; only manifests, scripts, and small smoke results are committed.

Tier	Purpose	Primary outputs
CI smoke	Install/regression checks on tiny fixtures	command success, output existence, stable summaries
Bulk matched	Main classical benchmark (ATAC vs. matched TF ChIP/CUT&RUN)	TFBS labels, predictions, AUROC/AUPRC, reports
Depth robustness	Shallow-depth sensitivity (50/25/10/5%)	performance-vs-depth and calibration curves
Motif removal	Motif-relaxed/free and de novo validation	recovered ChIP-supported sites, rediscovered-motif stats
Supervised TFBS	Tabular model benchmark	probability metrics, calibration, comparator tables
Variant	Translational benchmark (caQTL/allele-specific)	delta-score enrichment, case studies
Pseudobulk	Single-cell extension	cell-type TF programs, group comparisons

2.6. Reproducibility Engineering

The package ships a unit-test suite covering numerical helpers, CLI behavior, and deterministic golden CLI regressions for footprint scoring, differential binding, and aggregate plotting, with an opt-in slow regression for bias correction. A continuous-integration workflow runs the suite on a clean checkout using only small fixtures. Every benchmark result is expected to store random seeds, tool versions, command lines, and the resolved manifest. Figures are saved as vector PDF/SVG plus 300-dpi PNG previews, with source plotting tables retained for auditability.

3. Results

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the four core ATAC footprinting workflows behind a single, consistent command-line interface with shared region/signal conventions, so a complete analysis—bias correction, footprint scoring, differential binding, and aggregate visualization—runs without crossing package boundaries or reformatting intermediates. The optional YAML runner reproduces any pipeline from a declarative config, and the GUI exposes the same operations with run history and output discovery for less command-line-oriented users. Because every wrapper expands to the same underlying commands, results are identical across entry points, and the YAML config doubles as a machine-readable record of the analysis.

3.2. Deterministic Regression and CLI Coverage

The classical commands are covered by deterministic golden regressions that pin output summaries for footprint scoring, differential binding, and aggregate plotting, plus CLI smoke tests for all public and legacy entry points and YAML dry-runs. This converts "did this change alter results?" from a manual judgment into an automatically checked contract, and provides the stable base on which the opt-in modules are layered without disturbing default behavior.

3.3. Multiscale and Nucleosome-aware Extension

The multiscale module augments single-scale footprint scoring with a scale-by-position view of accessibility depletion. The compressed NPZ tensor sidecar serves both as a paper-figure input and as the substrate for the competition decomposition. On synthetic constructs, narrow TF-scale and broad nucleosome-scale depletions are recovered at their respective scales, and the companion aggregate visualization (scale-by-position heatmap, collapsed profile, and per-scale means) renders these directly. Crucially, default single-scale scoring is unchanged unless multiscale mode is requested.

3.4. Supervised, Motif-relaxed, De novo, and Variant Modules

The supervised TFBS path turns candidate sites into motif-centric feature tables and trains reproducible tabular models with saved bundles and calibration outputs, providing a labeled-data complement to classical scoring. The motif-relaxed/free path recovers candidate sites from signal alone or from lower-threshold motif merges and reranks them with motif-family and model evidence, enabling recovery of occupancy that strict motif thresholds miss. The de novo path orchestrates reproducible MEME/DREME and Tomtom command plans and summarizes results into TSV/HTML reports with inline consensus logos. The variant module annotates alleles with reference checks, candidate overlap, deterministic sequence-context deltas, optional PWM ref/alt motif deltas, and optional model probability deltas. Each module is independently tested and documented, and none alters the classical commands when unused.

3.5. Replicate-aware Uncertainty and Competition Decomposition

The replicate-aware report adds an interpretable uncertainty layer to differential binding: comparisons are categorized by replicate support, and effect sizes are reported with a p-value–derived standard error, a replicate-driven shrinkage weight, and confidence intervals, so single-replicate comparisons are flagged and conservatively shrunk relative to replicate-supported ones. The competition decomposition attributes overlapping footprint signal to TF-scale, nucleosome-scale, and shared (competing) components per region, with a competition index and a dominant-component label, surfacing loci where TF and nucleosome occupancy compete rather than treating accessibility as single-scale.

3.6. Benchmark and Figure Infrastructure

The tiered benchmark design (Table 1), the metric/calibration/bootstrap scripts, the label-overlap and motif-removal table builders, and the pipeline runner together provide an executable path from public ENCODE data to publication-ready metrics and figures. To demonstrate this path end-to-end on real data, we benchmarked four scoring paradigms on ENCODE ATAC-seq peaks restricted to chromosome 4, labeling each accessible region positive when it overlaps a matched ChIP-seq peak for the target factor. The compared predictors were raw accessibility magnitude, best JASPAR PWM motif log-odds score, cut-site footprint occupancy at the best motif hit where ATAC BAM data were available, and an integrated `fp-tools` score from out-of-fold logistic regression over accessibility, motif score, local GC content, and available footprint features. For tasks without BAM-derived cut-site footprints, the integrated model used the available accessibility, motif, and GC features. We evaluated 12 cell–TF tasks: six transcription factors in K562 (CTCF, GATA1, SPI1, REST, MAX, JUND; $n = 1{,}900$ accessible regions each), two in GM12878 (CTCF, SPI1; $n = 9{,}482$), and four in HepG2 (CTCF, MAX, JUND, REST; $n = 8{,}068$).

The integrated `fp-tools` score achieved the highest AUROC in all 12 tasks (Table 2, Figure 1). Mean AUROC across tasks was 0.52 for accessibility, 0.77 for motif-only scoring, and 0.80 for the integrated score. In K562 tasks with cut-site footprints, footprint-only signal averaged 0.55 AUROC: it was informative for CTCF but was not sufficient as a standalone predictor across factors. Motif-only scoring remained a strong baseline, especially for information-rich motifs such as CTCF and SPI1, but the integrated model consistently matched or improved it. The largest gains over motif-only scoring occurred for MAX and JUND, including K562 MAX (0.74 → 0.83), HepG2 MAX (0.64 → 0.75), K562 JUND (0.83 → 0.87), and HepG2 JUND (0.63 → 0.67). These results support the intended role of `fp-tools`: preserving interpretable classical predictors while providing a reproducible route to combine sequence, accessibility, composition, and footprint features when those data are available. Full AUROC/AUPRC values and bootstrap confidence intervals are reported in the benchmark metrics table distributed with the analysis outputs (see Data Availability).

Table 2. Integrated method-comparison benchmark on ENCODE ATAC-seq peaks (chromosome 4) labeled by matched ChIP-seq overlap. Values are AUROC point estimates; the full metrics TSV contains AUPRC and 200-resample bootstrap 95% CIs. The footprint-only column is shown only where BAM-derived cut-site footprints were computed.

Cell / TF	Pos. frac.	Access.	Motif	Footprint	Integrated
K562 / CTCF	0.36	0.53	0.87	0.68	**0.88**
K562 / GATA1	0.08	0.58	0.72	0.53	**0.73**
K562 / SPI1	0.28	0.47	0.79	0.54	**0.79**
K562 / REST	0.06	0.48	0.72	0.56	**0.73**
K562 / MAX	0.49	0.44	0.74	0.56	**0.83**
K562 / JUND	0.08	0.59	0.83	0.41	**0.87**
GM12878 / CTCF	0.26	0.57	0.90	–	**0.90**
GM12878 / SPI1	0.28	0.50	0.78	–	**0.78**
HepG2 / CTCF	0.28	0.59	0.86	–	**0.87**
HepG2 / MAX	0.35	0.49	0.64	–	**0.75**
HepG2 / JUND	0.13	0.49	0.63	–	**0.67**
HepG2 / REST	0.006	0.51	0.81	–	**0.86**

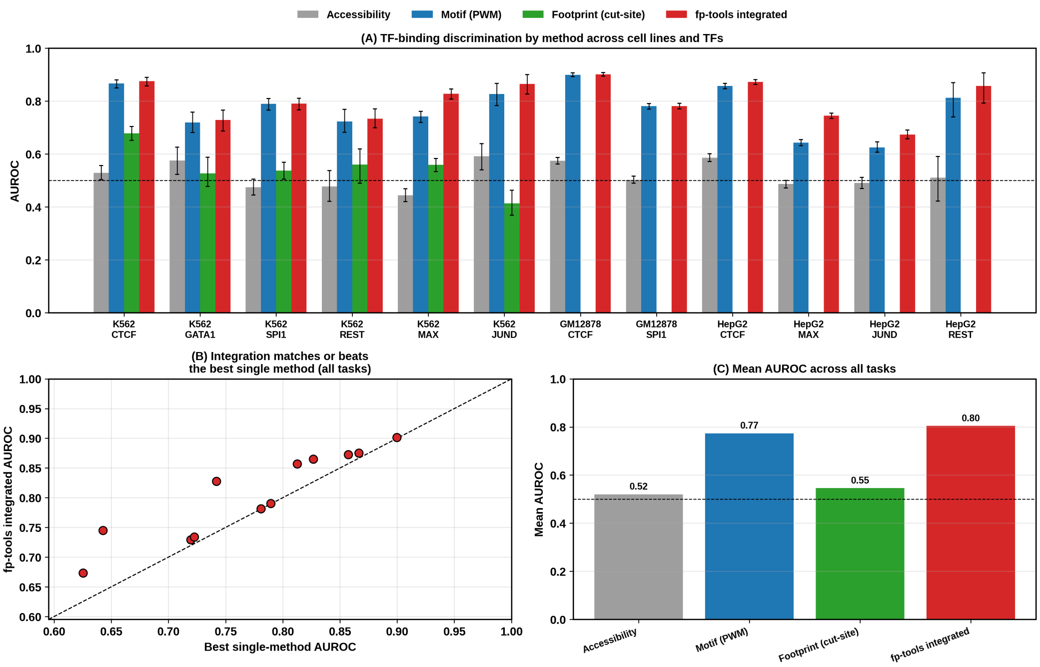

Figure 1. Multi-method benchmark across three ENCODE cell lines and 12 cell–TF tasks. Panel A compares per-task AUROC for raw accessibility, motif-only scoring, cut-site footprint scoring where available, and the integrated `fp-tools` model. Panel B compares the integrated model with the best single-method score for each task. Panel C summarizes mean AUROC by method. Generated by `paper/scripts/plot_method_comparison_multi.py`.

3.7. Footprinting Requires Cut-Site Resolution

Footprinting infers occupancy from a local depletion of Tn5 insertions at the bound site, flanked by hyper-accessible edges. We tested this directly on K562 CTCF (chromosome 4) by computing a footprint-occupancy score at each peak's best CTCF motif match—the mean signal in the flanks minus the mean signal at the motif site—using two different signal representations (Figure 2). On a smoothed fold-change-over-control coverage track the score is uninformative and in fact inverted (AUROC = 0.24), because coverage is *enriched*, not depleted, at bound accessible sites. When the same score is computed from base-resolution Tn5 cut sites derived from the ATAC BAM, it recovers a genuine footprint signal (AUROC = 0.68, 95% CI 0.65–0.70)—a large gain that arises purely from using the correct, cut-site-resolution signal that `fp-tools`' bias-correction and footprint-scoring pipeline is built to produce. For CTCF, whose long motif is already highly predictive, the cut-site footprint is an orthogonal but weaker line of evidence and does not exceed the motif score; footprinting is expected to contribute most where motifs are weak or where condition-specific occupancy changes are the question, which the differential binding and replicate-aware modules target.

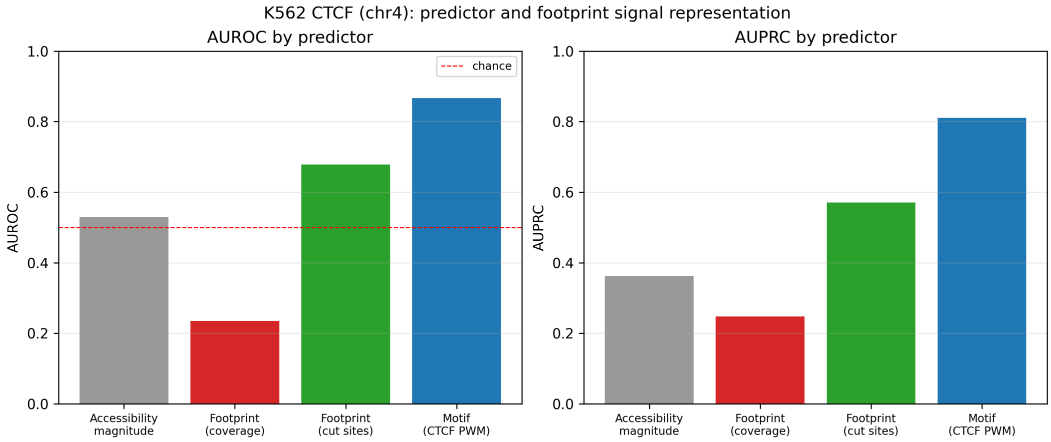

Figure 2. Predictor comparison for K562 CTCF (chromosome 4). A footprint-occupancy score is informative only at Tn5 cut-site resolution (green, AUROC 0.68); the same score on smoothed coverage is inverted and uninformative (red, 0.24). Raw accessibility (grey) is near chance and the CTCF motif (blue) is the single strongest predictor.

4. Discussion

`fp-tools` occupies a deliberate position in the regulatory-genomics tool landscape: rather than introducing a single new algorithm, it integrates established classical footprinting with modern multiscale, supervised, de novo, and variant-oriented capabilities inside one reproducible, testable package. Its comparative strengths are interface consistency, deterministic regression coverage, and a conservative extension model in which new science is opt-in and non-disruptive. These properties target precisely the friction that slows applied ATAC-seq analysis and cross-method benchmarking today.

We do not claim parity with deep sequence-to-profile models on tasks where learned base-resolution sequence features dominate; ChromBPNet-class models and their attribution-based interpreters remain state of the art for sequence-driven occupancy and variant-effect prediction [6]. `fp-tools` instead aims to make classical and tabular-supervised analyses reproducible, composable, and easy to benchmark against such models on shared labels. The multiscale, competition-decomposition, and replicate-uncertainty modules are pragmatic additions grounded in the available signal: in particular, the replicate-aware standard error is reconstructed from reported p-values rather than from per-replicate scores, so it should be read as a calibrated uncertainty heuristic rather than a full hierarchical model, and per-replicate inputs would enable a stronger formulation in future work.

Limitations include the dependence of the public-benchmark results on external data acquisition and comparator execution, which are intentionally kept outside version control for size and licensing reasons; the tabular supervised models trade base-resolution sequence modeling for transparency and speed; and several roadmap items (attribution-based discovery, RNA-prior weighting for pseudobulk, hierarchical differential binding) are scaffolded but not yet fully realized. These are natural directions for subsequent releases.

5. Conclusions

We presented `fp-tools`, a standalone, reproducible command-line platform that unifies classical ATAC-seq footprinting with opt-in multiscale, supervised, motif-discovery, variant-scoring, pseudobulk, replicate-uncertainty, and competition-decomposition modules. By keeping core workflows stable and test-covered while adding new scientific behavior as composable, independently validated modules—and by pairing them with a tiered public-data benchmark and figure infrastructure—the package lowers the barrier to standardized, benchmarkable regulatory-genomics analysis. Future work will complete the full public biological benchmark, strengthen the supervised and hierarchical differential-binding models, and integrate attribution-based discovery.

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Informed Consent Statement: Not applicable.

Data Availability Statement: `fp-tools` is openly available at https://github.com/oncologylab/fp-tools. All benchmark manifests, schemas, analysis scripts, and figure generators are included in the repository. Public input datasets are obtained from the ENCODE Project [7] and motif databases (e.g., JASPAR [8]) via the included discovery and download scripts; large raw and processed data and full benchmark outputs are regenerated from the committed manifests rather than stored in version control.

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1. Buenrostro, J.D.; Giresi, P.G.; Zaba, L.C.; Chang, H.Y.; Greenleaf, W.J. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nat. Methods* **2013**, *10*, 1213–1218.
2. Bentsen, M.; Goymann, P.; Schultheis, H.; Klee, K.; Petrova, A.; Wiegandt, R.; Fust, A.; Preussner, J.; Kuenne, C.; Braun, T.; et al. ATAC-seq footprinting unravels kinetics of transcription factor binding during zygotic genome activation. *Nat. Commun.* **2020**, *11*, 4267.
3. Li, Z.; Schulz, M.H.; Look, T.; Begemann, M.; Zenke, M.; Costa, I.G. Identification of transcription factor binding sites using ATAC-seq. *Genome Biol.* **2019**, *20*, 45.
4. Hu, Y.; et al. Multiscale footprinting reveals the regulatory grammar of chromatin accessibility (PRINT/scPrinter). *Nature/bioRxiv* **2024**.

5. Cazares, T.A.; Rizvi, F.W.; Iyer, B.; Chen, X.; Kotliar, M.; Bejjani, A.T.; Wayman, J.A.; Donmez, O.; Wronowski, B.; Parameswaran, S.; et al. maxATAC: Genome-scale transcription-factor binding prediction from ATAC-seq with deep neural networks. *PLoS Comput. Biol.* **2023**, *19*, e1010863.
6. Pampari, A.; et al. ChromBPNet: base-resolution models of chromatin accessibility reveal the sequence basis of TF binding and variant effects. *bioRxiv/Nature* **2024**.
7. The ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature* **2012**, *489*, 57–74.
8. Rauluseviciute, I.; et al. JASPAR 2026: update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.* **2026**.